ELSEVIER

Earth and Planetary Science Letters 133 (1995) 353–366

EPSL

A paleomagnetic study of Mesozoic sediments from the Junggar and Turfan basins, northwestern China

Jean-Pascal Cogné [a,*], Yan Chen [a], Vincent Courtillot [a], Frédéric Rocher [a], Gongque Wang [b], Meixiang Bai [c], Hongzi You [b]

[a] *Laboratoire de Géomagnétisme et Paléomagnétisme, Institut de Physique du Globe, 4 place Jussieu, 75252 Paris-Cedex 05, France*
[b] *Xinjiang Engineering Institute, Urumqi, People's Republic of China*
[c] *Bureau of Seismology of Xinjiang, Urumqi, People's Republic of China*

Received 7 November 1994; revised 28 April 1995; accepted 30 April 1995

Abstract

Mesozoic sedimentary rocks have been collected from the Karamay area (northwestern Junggar block) and in the Turfan basin (northeastern Tarim block) for a paleomagnetic study with the aim of testing the rigid nature of these blocks and to better constrain their paleogeographic relationships within central Asia prior to the collision with India. The results from the Karamay area are disappointing, Lower Cretaceous grey sandstones having apparently been recently remagnetized. This remagnetization is carried by magnetite, and yields highly clustered normal polarity directions. Both a medium (MTC) and a high (HTC) unblocking temperature component are isolated from flat-lying or weakly tilted Jurassic red sandstones. The MTC could be carried by maghemite and the HTC is carried by hematite. The two components are statistically identical, and seem to correspond to an ancient magnetization. However, there is no fold test and comparison with previous data from Junggar and the reference APWP of Siberia does not allow us to decide between a Middle Jurassic primary magnetization and an Upper Jurassic or Cretaceous remagnetization. Stronger constraints are obtained from Upper Jurassic to lower Eocene fine-grained red sandstones from anticlines that have been thrust southwards, in the Turfan area of South Tien Shan. After the elimination of a recent overprint that is probably related to weathering, a high coercivity, high unblocking temperature component carried essentially by hematite is isolated. All directions from the Jurassic to the lower Eocene, which yield positive fold and reversal tests, are statistically indistinguishable. This is compatible with a primary origin for these magnetizations, similar to what has previously been found in Junggar, Tarim and western Tibet. The corresponding Turfan pole is not statistically distinct from that of Tarim for the same period, in accordance with the notion of a large, quasi-rigid Tarim block remaining attached to Eurasia in a rather constant position and at a constant latitude throughout the second half of the Mesozoic. This new datum can therefore be included to derive a new mean pole for the Tarim that is valid for the period from the Middle Jurassic to the end of the Cretaceous: 68.4° N, 224.5° E ($A_{95} = 3.1°$).

1. Introduction

The many paleomagnetic studies that have been performed in China during the last decade have improved our knowledge of the accretion history of the various Asian blocks [1] and of the tectonic

* Phone (33 1) 44 27 60 93; fax (33 1) 44 27 74 63; e-mail cogne@ipgp.jussieu.fr

0012-821X/95/$09.50 © 1995 Elsevier Science B.V. All rights reserved
SSDI 0012-821X(95)00083-6

modifications of this assemblage under the effect of the collision of India [2–5].

Recent paleomagnetic studies of Mesozoic formations in Tibet [6–10], the Tarim basin [11,12] and the Junggar basin [13] led to a model of continental deformation for central Asia [4]. The Cretaceous is the period for which the largest and best quality data are available. It was found that the total amount of convergence between India and Siberia since the time of collision of India is largely distributed within the mountain ranges (Himalaya, Kunlun, Tien Shan, Altai) which separate the different blocks. This interpretation of course depends on the definition of the relative paleopositions of the different blocks for different epochs prior to the collision and thus on the number of paleomagnetic pole determinations within each block. At present, only the Tibetan Plateau is quite well sampled, whereas the Cretaceous pole for the Tarim is defined only in its western [12] and northwestern [11] parts, and that for Junggar only on its southern border, in the northern piedmont of the Tien Shan range [13].

This paper reports new paleomagnetic results from Mesozoic continental sediments (Middle Jurassic to Early Tertiary) of the Tarim and Junggar collected in two distinct areas, the Turfan basin located on the southern border of the Tien Shan (east of Urumqi) and in the Karamay area of northwestern Junggar (Fig. 1).

2. Sampling

During a field trip in September 1991 we collected samples from a total of 44 sites around the Junggar basin and the Tien Shan area. Generally eight to ten samples were drilled and oriented in-situ by using sun and magnetic compasses. Two main collections, totalling 37 sites, were sampled in two specific areas (Fig. 1):

Turfan area (South Tien Shan): Mesozoic sediments were collected from a series of anticlines thrust to the south that border the southern piedmont of the Tien Shan range, east of the Turfan basin. Three formations (ages and formation names follow the 1/200,000 geological map) of fine-grained red sandstones were sampled: two sites (No. 11 and 12) in the Upper Jurassic (J_{3q}) Qigu Formation, thirteen sites (No. 6–10 and 13–20) in the Lower Cretaceous (K_{1t}) Tugulu Formation, and four sites (No. 2–5) in the Upper Cretaceous–lower Eocene (K_2–E_1) Shanshan Formation. Sites were sampled in N–S valleys cross-cutting the roughly E–W trending anticlines: Sites 2–5 are situated at 43° N, 90.5° E, sites 6–12 at 42.8° N, 90° E, and sites 13–20 are located at 42.8° N, 89.5° E. The distance between the easternmost and westernmost valleys is thus about 100 km. Because the anticlines are faulted and overthrust to the south, most of the sites are situated in normal limbs, dipping to the north, and only two sites (19 and 20 in K_{1t}) could be found with beds dipping to the south.

Karamay area (northwestern Junggar): Here, depending on their age and lithological nature, two collections are distinguished: (1) A set of seven sites (21–26, situated at 46° N, 86° E, and 37 situated at 45.6° N, 84.9° E) was sampled in the Lower Cretaceous (K_{1t}) Tugulu Formation. These consist of flat-lying light yellow to grey sandstone beds. (2) The other set of eleven sites (27–36 and 38, situated at 45.6° N, 84.9° E) was sampled in the Middle Jurassic (J_{2x-s}) gently dipping dark red sandstones of the Xishanyao and Sangonghe Formations. Note that although the beds are nearly horizontal in both formations, the underlying Jurassic beds show generally greater dips than the Cretaceous beds (Table 1). This denotes an unconformity between the Lower–Middle Jurassic strata and the overlying Upper Jurassic and Cretaceous. This unconformity is thought to arise from weak tectonic activity resulting in normal fault development during sedimentation, which affects the Triassic and Lower–Middle Jurassic formations [14–17]. In this area, there has been no further tectonic activity since the Late Jurassic, as shown by the flat-lying Cretaceous beds.

Samples from three of the 37 sites (3, 6 and 17) were accidentally destroyed (crushed) during transportation.

3. Paleomagnetic analysis and results

Magnetic declinations, computed using the sun and magnetic azimuths, average 1.6° W ($\pm 1.2°$, $n = 32$) in the Turfan area, and 4.3° W ($\pm 1.3°$, $n = 55$) in the Karamay region. All the orientations given below are corrected for these magnetic declinations.

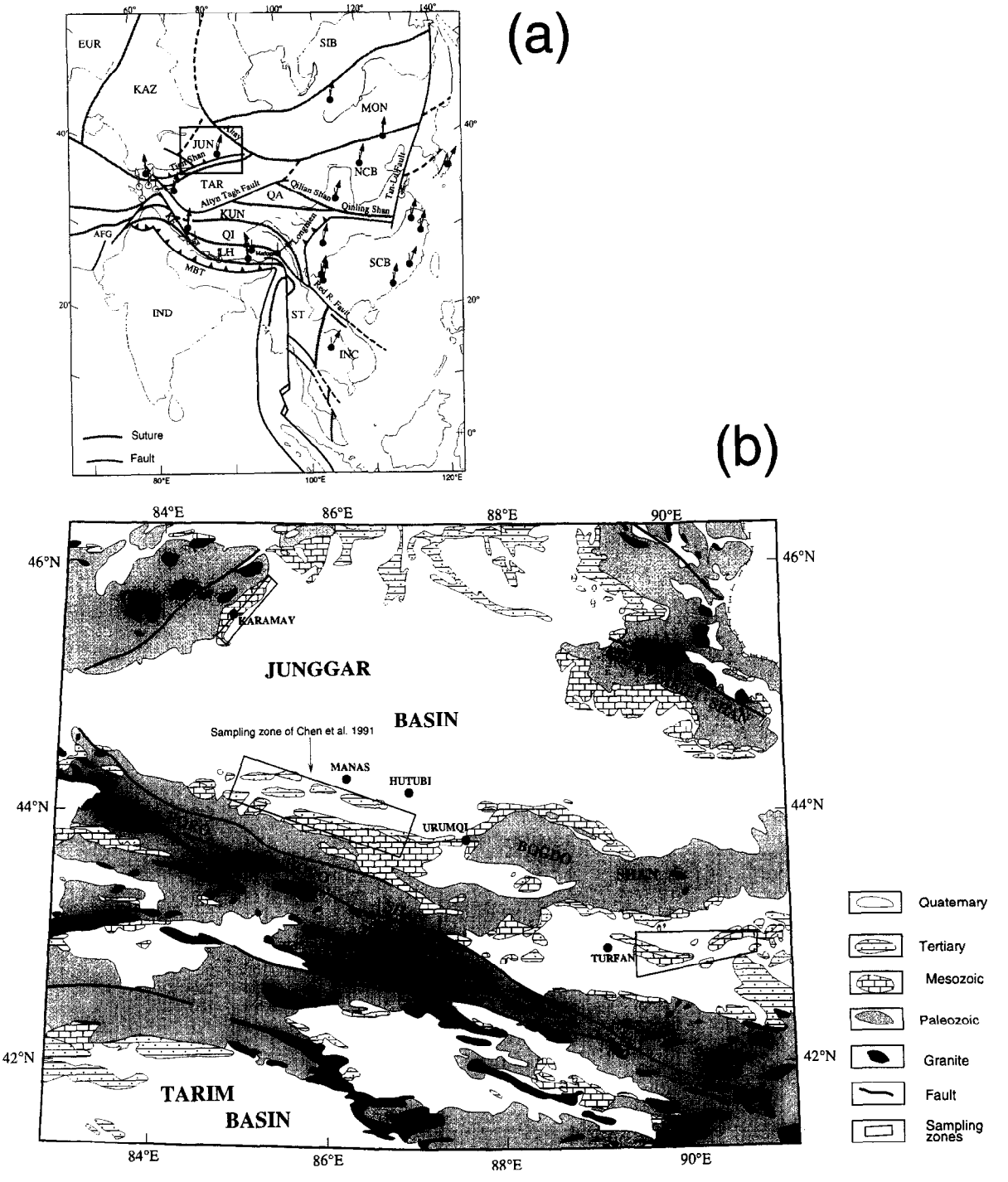


Fig. 1. (a) Schematic block map of China and Southeast Asia showing the main sutures and faults (*KF* = Karakorum Fault; *MBT* = Main Boundary Thrust). Major blocks: *AFG* = Afghanistan; *EUR* = Europe; *INC* = Indochina; *IND* = India; *IR* = Iran; *JUN* = Junggar; *KAZ* = Kazahkstan; *KUN* = Kunlun; *MON* = Mongolia; *NCB* = North China Block; *QA* = Qaidam; *QI* = Qiangtang; *SIB* = Siberia; *SCB* = South China Block; *ST* = Shan Thai; *TAR* = Tarim. Also indicated are the main locations where Cretaceous paleomagnetic data are available. Declination is indicated by arrows (after [4]). (b) Schematic geological map, after [17], showing location of the Karamay (northeastern Junggar) and the Turfan (northeastern Tarim) sampling areas and the location of the previous study of [13] (east of Urumqi).

Table 1
Site-mean paleomagnetic directions for the high temperature components

Site	n/N	Strike/Dip	D_g	I_g	D_s	I_s	k	α_{95}
Turfan area								
Upper Cretaceous (K_2-E_1)								
2N(*)	5/10	247.5/48	157.7	65.2	337.3	66.8	34	13.3 (*f*)
2R	4/10		229.1	-52.3	196.5	-22.7	180	6.9 (*f*)
4	6/6	238.5/50	87.3	58.5	11.5	47.8	132	6.0 (*m*)
5	3/4	238.5/48	258.9	-43.1	212.4	-40.1	222	9.7 (*m*)
Lower Cretaceous (K_1)								
7	4/7	289.5/56	68.0	-68.4	176.2	-46.2	27	18.0 (*f*)
9(*)	9/9	297.5/46	333.5	-74.2	233.7	-48.1	78	5.9 (*m*)
10	4/6	297.5/55	220.2	-75.9	210.8	-21.2	48	13.4 (*f*)
13	6/6	316.5/34	170.5	-57.0	194.5	-31.7	14	18.6 (*m*)
14	9/9	311.5/32	241.8	7.2	246.1	37.0	*(mean plane)*	
15	5/6	296.5/24	116.8	18.7	109.0	17.1	*(mean plane)*	
16	4/6	306.5/39	358.7	76.6	25.8	39.9	54	12.6 (*f*)
18	4/6	306.5/40	134.6	-71.7	190.7	-44.3	54	14.7 (*m*)
19	7/7	138.5/50	29.4	6.8	15.9	52.4	67	7.4 (*f*)
20(*)	5/6	152.5/70	18.8	37.7	299.0	47.5	80	8.9 (*m*)
Upper Jurassic (J3q)								
11	5/5	287.5/48	168.1	74.2	30.9	54.6	44	12.4 (*m*)
12	7/7	316.5/30	358.7	63.6	21.5	38.7	39	9.8 (*f*)
Mean								
	13/15		26.6	71.3	-	-	7	16.8 (*m*)
			-	-	17.9	40.3	37	7.0 (*m*)
Karamay area								
Lower Cretaceous (K_{1t})								
21	4/6	274.3/ 3	359.9	75.4	0.5	72.4	211	6.3 (*f*)
22	6/6	horizontal	2.9	69.5	2.9	69.5	103	6.6 (*f*)
23	5/5	244.3/ 4	350.2	71.5	347.3	67.7	471	3.5 (*f*)
24	5/6	328.3/ 7	353.8	68.2	10.1	66.4	257	4.8 (*f*)
25	4/6	horizontal	2.1	73.2	2.1	73.2	532	4.2 (*f*)
26	5/5	horizontal	7.7	58.2	7.7	58.2	188	5.6 (*f*)
37(*)	5/5	56.3/ 4	197.8	-24.6	199.4	-26.9	82	10.6 (*m*)
Mean								
	6/7		0.1	69.5	-	-	157	5.4 (*f*)
			-	-	2.3	68.1	169	5.2 (*f*)
Middle Jurassic ($J2_{x-s}$)								
27	6/8	91.3/12	13.6	48.3	18.1	59.9	54	9.2 (*f*)
28	5/6	97.3/16	16.6	55.1	24.3	70.6	62	10.5 (*m*)
29	11/11	35.3/10	15.7	52.8	29.2	54.9	225	3.0 (*f*)
30	7/7	17.3/14	18.3	59.2	40.1	56.0	78	6.9 (*f*)
31	4/6	84.3/ 2	17.3	50.0	18.4	51.8	111	9.3 (*m*)
32	4/4	78.3/14	353.9	61.4	359.1	75.2	99	9.3 (*f*)
33	6/7	horizontal	204.8	-47.4	204.8	-47.4	231	4.5 (*m*)
34	7/7	38.3/ 4	6.0	57.0	11.7	58.9	130	5.3 (*f*)
35	7/7	39.3/ 2	19.7	46.4	21.7	46.9	554	2.6 (*f*)
36	5/6	138.3/ 7	359.6	62.9	347.5	67.1	80	8.6 (*f*)
38	4/4	horizontal	16.9	52.5	16.9	52.5	168	7.1 (*f*)
Mean								
	11/11		13.9	54.2	-	-	112	4.3 (*f*)
			-	-	19.4	58.9	51	6.5 (*f*)

n/N = Number of entry in the statistics/number of treated specimens. Bedding is given as strike (direction of the horizontal of the plane) and dip (dip angle in the direction of strike +90°). D_g, I_g/D_s, I_s = Mean directions in geographic/in-situ (or stratigraphic, tilt-corrected) coordinates. k, α_{95} = Parameters of either Fisher [20] (*f*) or McFadden and McElhinny [21] (*m*) statistics. Sites marked with (*) are rejected from the final means. Data for sites 14 and 15 are the mean poles of the remagnetization planes.

Standard 2.5 cm diameter minicores were cut in 2.2 cm long specimens in the laboratory. All the magnetic measurements and demagnetizations were performed in the magnetically shielded room of the Paleomagnetic Laboratory at the *Institut de Physique du Globe de Paris / Université de Paris 7*. Remanent magnetization was measured with a three-component CTF cryogenic magnetometer. Most of the specimens were thermally demagnetized in ten to fifteen steps in a laboratory-built furnace. Magnetic changes during demagnetization were monitored by measuring bulk susceptibility after each heating step, using a Bartington susceptibility meter. The demagnetization results are plotted as orthogonal vector diagrams [18] and equal-area projections. Paleomagnetic directions or planes were determined using principal component analysis [19] and site mean directions were determined using Fisherian statistics [20], or McFadden and McElhinny [21] statistics in the case of combined directional data and remagnetization circles. At two sites, mean remagnetization planes were determined using Mardian [22] statistics. Isothermal remanent magnetization (IRM) acquisition experiments and thermal demagnetization of IRM were performed on a few selected samples, in order to help in the identification of the magnetic carriers. A

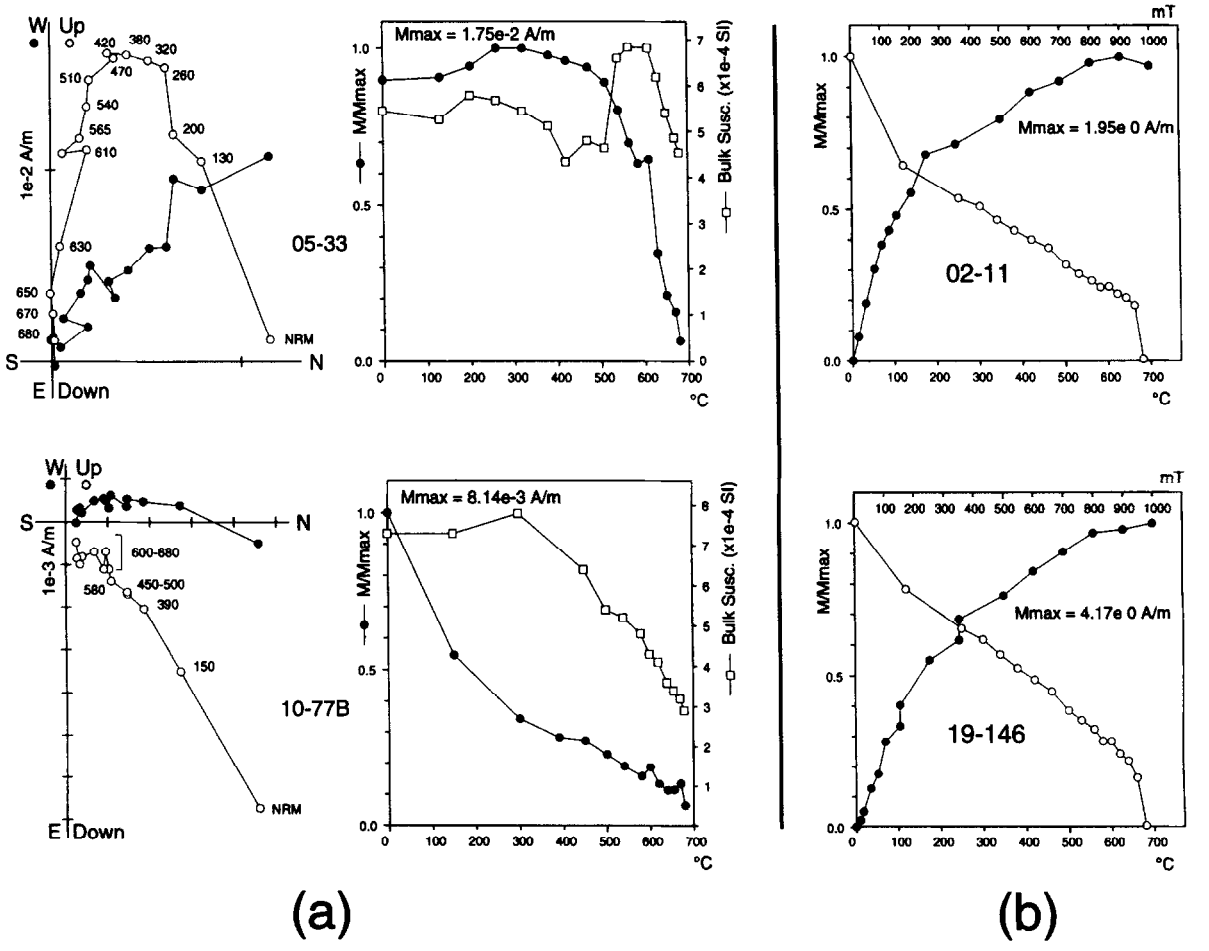


Fig. 2. (a) Typical thermal demagnetization curves of samples from the Turfan area. Left: Orthogonal vector plots [18] in in-situ coordinates; closed/open symbols = projection onto the horizontal/vertical plane; temperature steps are indicated in ° C. Middle: Curves of normalized intensity (●) and bulk susceptibility (□) in SI units vs. temperature. (b) Normalized IRM acquisition curves (●, upper scale) and normalized intensity decay curves of IRM during thermal demagnetization (○, lower scale).

total of 205 specimens from 34 sites were measured and demagnetized. Thirty out of these 34 sites gave the consistent results which are described below.

3.1. Turfan J_{3q} to K_2-E_1 formations

Typical orthogonal plots, natural remanent magnetization (NRM) intensity plots and susceptibility plots of thermal demagnetizations from these redbeds are shown in Fig. 2. In general, thermal demagnetization allowed the separation of two magnetization components. A low unblocking temperature component (LTC), with northerly declinations and downwards intermediate inclinations in geographic coordinates (i.e., before tilt correction), is removed below 300–400° C. The contribution of this component to the total NRM is variable but may represent more than 70% of the total intensity. The Fisherian average for this component is $D_g = 2.5°$, $I_g = 61°$, $k_g = 29$, $\alpha_{95} = 3°$ ($k_g/k_s = 4.8$) for 79 specimens before tilt correction, close to the present-day dipole field direction of $I = 62°$ at this location. This component is most likely a recent overprint.

After removal of this LTC, a high temperature component (HTC) which was unblocked by 680° C (Fig. 2a) could be identified in most of the samples. This HTC is either normal (i.e., with downward inclinations (e.g., specimen 10-77B in Fig. 2a)) or reverse (upward inclinations, see specimen 05-33 in Fig. 2a). Due to the very high unblocking temperatures of the HTC, and to the red colour of the beds, we assume that this component is carried by hematite. As may be seen, for example in specimen 05-33, the HTC does not always converge towards the origin of the orthogonal plots in the range 400–650° C. This may be due either to bad separation from the LTC, or to the presence of an overprint which overlaps the HTC in this temperature range. Above 630–640° C, mineralogical changes due to heating (as may be seen from changes in bulk susceptibility with temperature, Fig. 2a) may induce some noise in the determination of the final direction. (In that case a remagnetization great circle [23] is determined within the 400–650° C temperature range.)

IRM acquisition curves (Fig. 2b) show a continuous increase in intensity up to an applied field of 1 T where saturation is not reached. Thermal demagnetization of IRM acquired at 1 T shows a fairly large decrease in intensity for temperatures lower than 250° C, followed by a regular decrease and a sharp

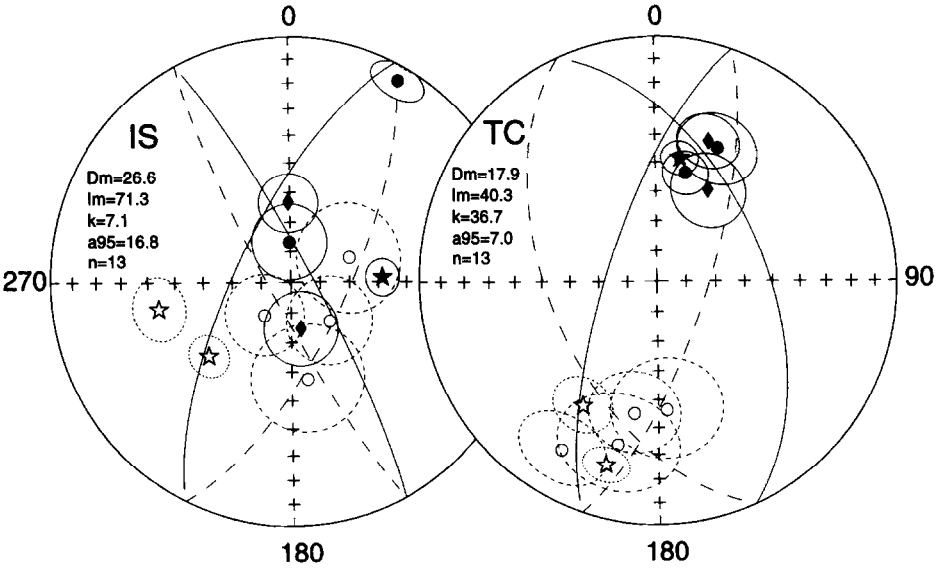


Fig. 3. Equal-area projections of site-mean directions of the high temperature components (HTC) with their α_{95} circles of confidence for the Turfan Mesozoic samples shown in-situ (IS, left) and tilt-corrected (TC, right). Closed/open symbols = downward/upward inclinations: Diamonds = Upper Jurassic (J_{3q}) data; circles = Lower Cretaceous (K_1) data; stars = Upper Cretaceous-lower Eocene (K_2-E_1) data. Great circles are the average remagnetization planes for sites 14 and 15.

breakdown between 660 and 680° C. Both thermal demagnetization of NRM and IRM experiments show that we have at least two magnetic carriers: (1) a high IRM coercivity–low unblocking temperature mineral which carries the present-day field LTC overprint, and which could probably be a hydroxide (goethite) of weathering origin, and (2) a high IRM coercivity–high unblocking temperature mineral which carries the HTC and is undoubtedly hematite. Although the presence of magnetite cannot be ruled out from these experiments, its contribution to the NRM does not appear to be significant.

All sites but No. 8 provided consistent HTC, or remagnetization planes, of either normal or reverse polarity. Site-mean directions for the HTC are listed in Table 1 and illustrated on equal-area projections in Fig. 3. For sites 14 and 15, only remagnetization planes could be determined, and because these planes were parallel to each other within each site, an average remagnetization plane had to be computed at

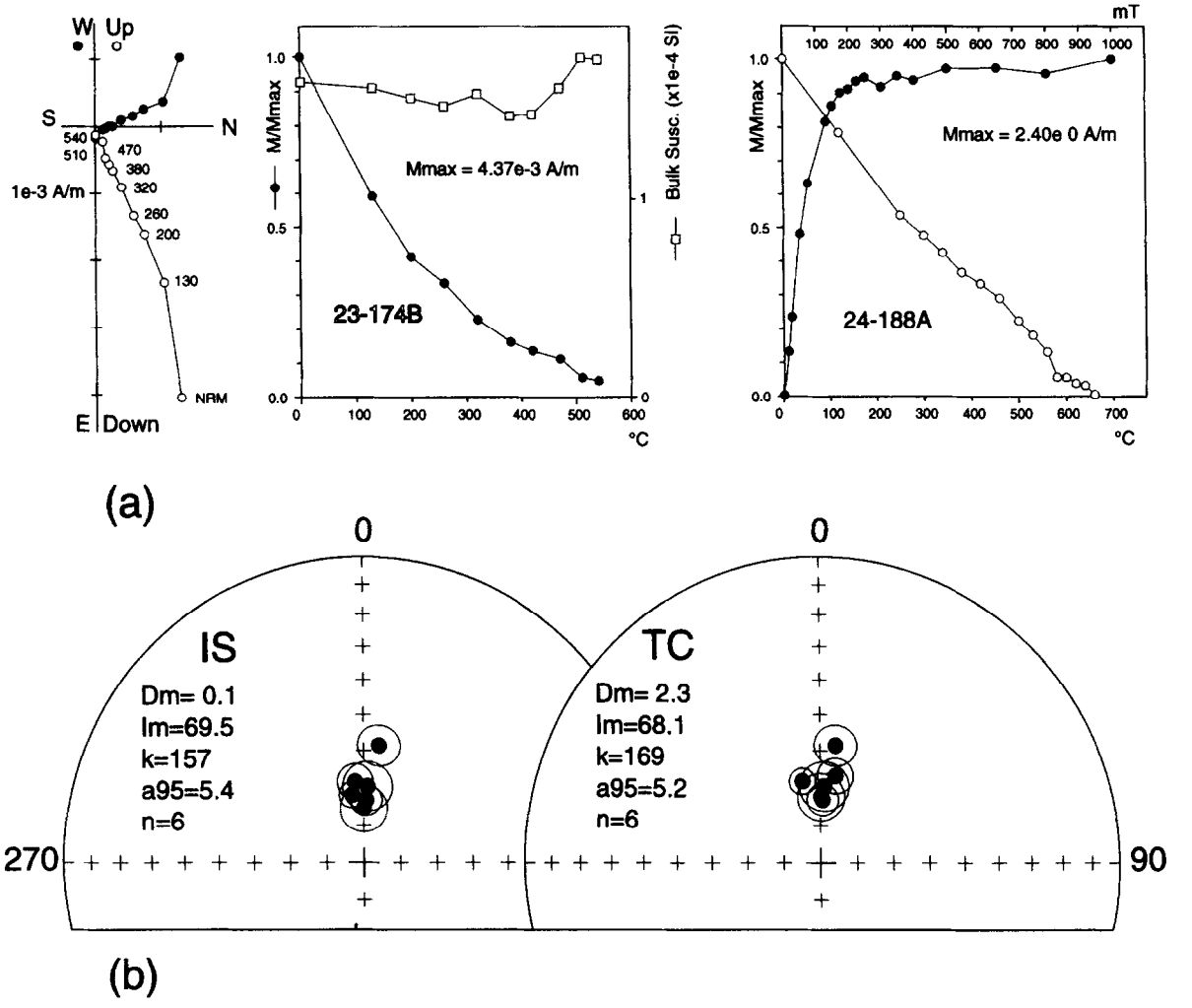


Fig. 4. (a) Typical thermal demagnetization curves of Lower Cretaceous (K_{1t}) Tugulu Formation samples from the Karamay area. Left: Orthogonal vector plots. Middle: Normalized intensity and bulk susceptibility curves. Right: IRM experiments. Same conventions as in Fig. 2. (b) Equal-area projections of site-mean directions of HTC with their α_{95} circles of confidence in-situ (IS, left) and tilt-corrected (TC, right). Same conventions as in Fig. 3.

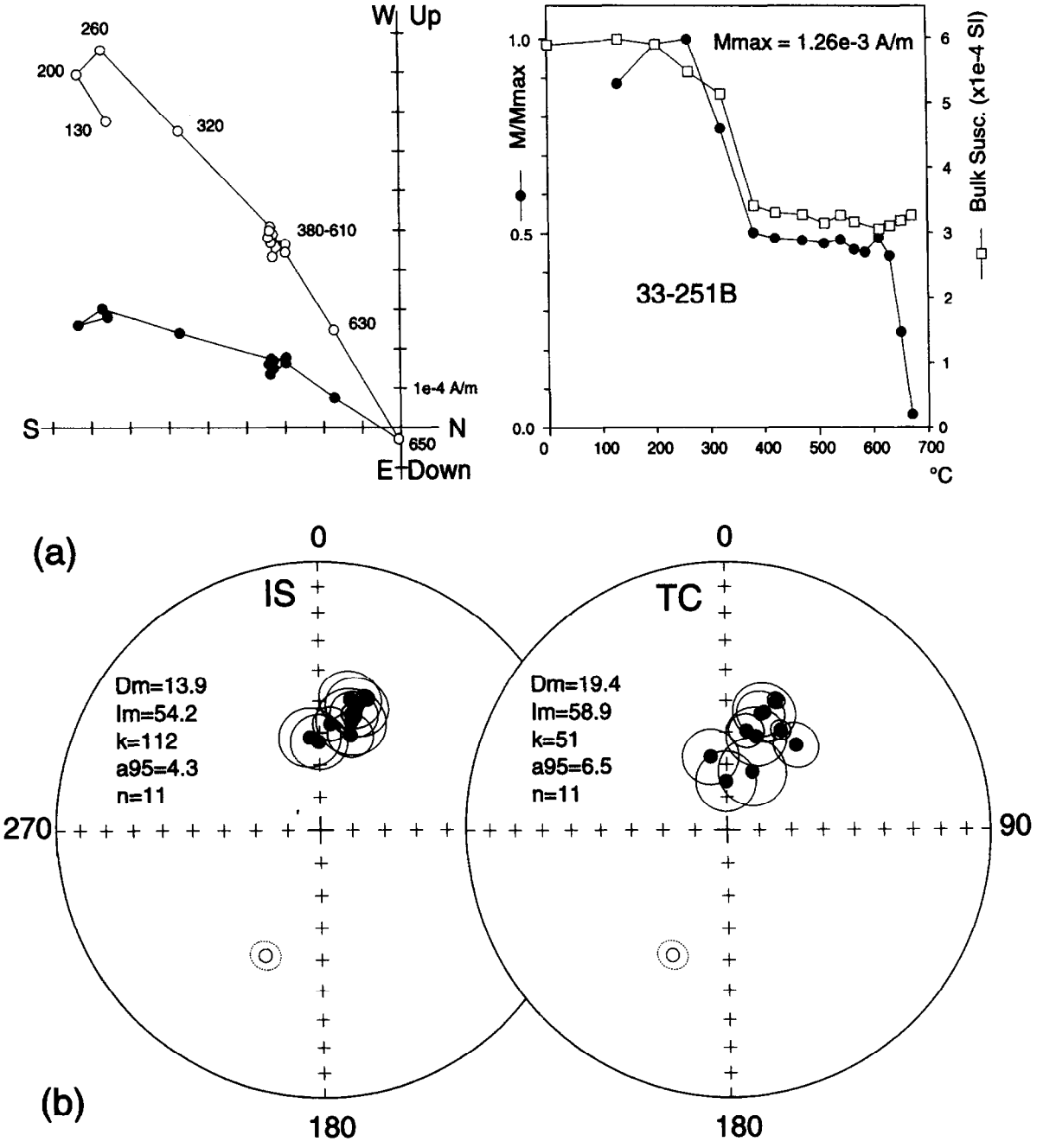


Fig. 5. (a) Typical thermal demagnetization curves of Jurassic (J_{2x-s}) Xishanyao and Sangonghe Formation samples from the Karamay area. Left: Orthogonal vector plots. Right: Normalized intensity and bulk susceptibility curves. Same conventions as in Fig. 2. (b) Equal-area projections of site-mean directions of HTC with their α_{95} circles of confidence shown in-situ (IS, left) and tilt-corrected (TC, right). Same conventions as in Fig. 3.

each. Another problem appeared with the site 2 data, where normal and reverse directions are not antiparallel. We have no definite explanation for this, except perhaps a bad separation of magnetization components in the case of the normal directions. Only reverse directions from site 2 are thus included in the final mean, computed using the statistics of McFadden and McElhinny [21] for a mixed population of vectors and great circles.

As may be seen from the data of Table 1 and Fig. 3, the Upper Jurassic (J_{3q}, diamonds in Fig. 3), Lower Cretaceous (K_{1t}, circles) and Upper Cretaceous–lower Eocene (K_2–E_1, stars) directions are not significantly different (after tilt correction), which is similar to the results in several of our earlier studies in central Asia [10,12,13]. We therefore computed a single mean for the whole population spanning from Upper Jurassic to lower Eocene. It is obvious that directions cluster upon unfolding, and the fold test [24] is positive at the 99% probability level, with $k_s/k_g = 5.3$, $n = 13$ (or $k_s/k_g = 7.2$, $n = 11$ using only HTC site-mean directions without the mean planes of sites 14 and 15). Although inclinations from reverse directions appear slightly more gentle than those from normal directions, the reversal test [25] is also positive. We therefore assume that the HTC, which is pre-tectonic in age, does correspond to the primary magnetization.

3.2. Karamay

Lower Cretaceous (K_{1t}) Tugulu Formation

The results of thermal demagnetization of NRM and the IRM experiments are shown in Fig. 4a. Apart from a viscous component removed by 100–150° C, which represents generally 50–70% of the NRM intensity, samples from this formation show a univectorial decay of magnetization up to about 540° C, where more than 90% of magnetization is lost. Bulk suceptibility (Fig. 4a, middle) remains stable during the thermal experiments. IRM acquisition curves (Fig. 4a, right) exhibit a rapid increase in magnetization intensity in fields up to 250 mT, followed by a slight increase up to 1 T. Thermal demagnetization of IRM shows the same type of behaviour as for NRM, except that about 5% of the intensity remains above 560° C and is demagnetized by 650° C. Therefore, there is probably a small amount of hematite within these rocks. We assume that the high IRM coercivity–low unblocking temperature mineral carries the viscous component, and that the characteristic remanent magnetization (ChRM) component is probably carried by magnetite, due to its low IRM coercivity and its 200–540° C unblocking temperature range.

Site-mean directions for the ChRM are given in Table 1 and illustrated in Fig. 4b. The data show very good clustering at the site scale, with k parameters in the 100–500 range. All ChRM directions from sites 21–26 exhibit northerly downwards directions, while those from site 37 have southerly upwards directions, but are not antiparallel to the other population. We thus do not consider site 37 as pertaining to the same population, and its direction is rejected from the final mean. Because the beds are flat-lying, no fold test can be performed for these data, but we note that both in-situ and tilt-corrected formation mean directions contain the present-day dipole field (PDF) direction ($D = 0°$, $I = 64°$) within their α_{95} cone of confidence. Because of high within-site cluster, exclusive normal directions, and a mean direction that contains the PDF, we interpret the ChRM in the Cretaceous grey sandstones of sites 21–26 from the Karamay area as a recent, or present, remagnetization.

Jurassic ($J2_{x-s}$) Xishanyao and Sangonghe Formations

The thermal demagnetization curves are shown in Fig. 5a. There is a strong (up to 95% of total NRM intensity) low temperature component which is removed by 100–150° C. This LTC is followed by a medium temperature component (MTC) which unblocks between 250 and 400° C. This is particularly obvious in specimen 33-251B (Fig. 5a) where it represents 50% of the NRM remaining after 130° C. Such a good definition of the MTC is however not the general case, and it could be determined in only 37 specimens out of the 73 analyzed. The susceptibility curve of specimen 33-251B (Fig. 5a, right) shows a sharp drop between 320 and 380° C: the MTC may be related to the destabilization of some magnetic phase, although the unblocking temperature of some mineral may have been reached at the same time. Two magnetic phases show such breakdown at intermediate temperatures: pyrrhotite and maghemite.

However, pyrrhotite usually destabilizes to magnetite upon heating, bulk susceptibility increases [26–28] and this often results in the acquisition of a spurious magnetization on cooling down, whereas maghemite is usually converted to low susceptibility hematite [29,30]. Because of the decrease in susceptibility observed here, maghemite could be the carrier of MTC in our samples. After removal of the MTC, when present, magnetization remains stable up to temperatures higher than 610° C, followed by a breakdown at 630–650° C, defining a HTC carried by hematite.

Because MTC directions were determined in too small a number of specimens, we computed a mean direction at the specimen level. The mean direction of the MTC is $D_g = 14.1°$, $I_g = 56.8°$ ($k = 43$, $\alpha_{95} = 3.6°$) before tilt correction, and $D_s = 20.2°$, $I_s = 61.1°$ ($k = 34$, $\alpha_{95} = 4.1°$) after correction for bedding tilt ($n = 37$). The fold test [24] is thus inconclusive at the 95% probability level.

Site mean directions of the HTC are given in Table 1 and illustrated in Fig. 5b, from which we make the following observations:

(1) All specimens from site 33, and two specimens from site 28, show reverse directions which are antiparallel to the normal ones.

(2) Because the beds have small dips (15°), the tilt correction has little effect on the distribution of the paleomagnetic vectors. However the k parameter decreases from 112 to 51 upon untilting. This is not significant at the 99% probability level but this decrease is significant at the 95% probability level, following the fold test of McElhinny [24]. Other tests, including Tauxe and Watson's [34] and McFadden and Jones' [35], appear to be either negative or inconclusive at the 95% probability level. We therefore discuss both in-situ and tilt-corrected directions in the following section. Note, however, that the mean directions of the HTC, as well as of the MTC, in either in-situ or tilt-corrected coordinates, do not contain the PDF direction ($D = 0°$, $I = 64°$) within their α_{95} cone of confidence.

(3) Finally, we note that the mean directions of MTC understood to be carried by maghemite, and of HTC understood to be carried by hematite, are identical at the formation level.

Owing to the fact that two different minerals carry the same magnetic component, one could conclude that these beds are remagnetized. However, such a remagnetization would not be likely to be due to weathering, because of the nature of the magnetic carriers, or to be recent because the mean direction is different from the PDF. There is no field or petrographic evidence of heating of these beds or of fluid circulation within them, and, moreover, these flat-lying beds have not been strongly tectonized (folded). Therefore thermal or thermochemical remagnetization during a tectonic event appears unlikely. Maghemite could have been inherited by transportation from the surrounding relief during deposition and could carry a detrital or post-detrital magnetization (DRM or PDRM). Hematite, which generally forms by dehydration of hydroxydes or of silicates [29,31] during compaction, would have blocked a PDRM with a short time delay with respect to maghemite. This could explain the slight angle between MTC and HTC as observed in the demagnetization curves, although these are not distinguishable at the formation scale. Having no strong evidence for the age of this magnetization, we are left with the need to compare the corresponding pole with reference poles for various ages from Junggar or neighbouring Siberia. This is done in the following discussion.

4. Discussion

4.1. Turfan

As shown above, the HTCs from the Upper Jurassic, Lower Cretaceous and Upper Cretaceous–lower Eocene beds of the Turfan area yield a positive fold test at the 99% level, with dual polarity, and are distributed as a single population spanning all this time. Assuming these are primary magnetizations, this would indicate stability of the magnetic field or little paleogeographic change from the Upper Jurassic to the Lower Tertiary. The stability of Cretaceous virtual geomagnetic poles (VGP) for Eurasia has already been demonstrated through the analysis of the Eurasian apparent polar wander path (APWP) by Besse and Courtillot [32], and locally verified through paleomagnetic studies of Mesozoic sediments from Junggar [13], Tarim [11,12] and western Tibet [10].

We therefore compute a single pole for the three

sampled age groups, using the tilt-corrected average directions of Table 1, at the mean location 43° N, 90° E. This pole is given in Table 2, and illustrated in Fig. 6a where it is compared to the Cretaceous poles from Junggar and Tarim [12,13], and to the APWP of Eurasia and a mean Cretaceous (65–130 Ma) VGP from Europe [32].

The mean Cretaceous VGP for the Turfan basin clearly appears closer to that of Tarim (with a non-significant angular difference of $4.3 \pm 6.2°$) than to those of Junggar (angular difference of $8.4 \pm 6.7°$) and Eurasia (65–130 Ma pole, angular difference of $13.7 \pm 5.5°$). This implies that no (or slight) relative movement has taken place between Tarim and Turfan since the Late Jurassic, which suggests a rigidity of the Tarim block since that time; relative movement might have occurred between Junggar and Turfan, with no significant rotation ($4.0 \pm 6.7°$), but possibly with some latitudinal movement ($6.4 \pm 6.7°$). This analysis significantly extends the geographical coverage of the rigid Tarim block (see Fig. 1) and allows us to compute a new mean Cretaceous VGP for it, including the results obtained in the Turfan basin. This new Tarim VGP (Table 2) differs only by 0.7° from the previously published pole [12], and the conclusions reached in the latter paper remain essentially unchanged, as well as our previous interpretation of the continental deformation of central Asia [4,10,12,13,33].

4.2. Karamay formations

We have already found that the magnetization of the Lower Cretaceous (K_{1t}) Tugulu Formation is most likely a recent remagnetization. We now return to results from the Jurassic (J_{2x-s}) Xishanyao and Sangonghe Formations.

As discussed above, the fold test for the HTC is negative at the 95% probability level, and inconclusive at the 99% level. It is therefore difficult to assess the primary origin of this magnetization and we discuss the possible significance of poles com-

Table 2
Virtual geomagnetic poles of this study, and compilation of Cretaceous VGPs of the Tarim and Junggar basins

Area	Age	N	S Lat.	S Long.	P Lat.	P Long.	dp/dm	Reference
Tarim								
Yingjisha	Ku	6	38.5	76.4	71.0	234.0	6.8/11.6	[12]
Yingjisha	Kl	3	38.5	76.4	70.4	212.1	6.6/10.8	"
Wuqia	Ku	11	39.5	75.0	70.8	222.6	5.4/8.9	"
Wuqia	Kl	7	39.5	75.0	66.3	226.6	9.0/15.9	"
Kuche	Ku	4	41.6	83.5	66.2	223.3	6.1/10.3	[11]
Average (a)	K	5	-	-	69.1	223.8	A_{95}=3.5	[12]
Turfan	K	13	43.0	90.0	65.1	227.8	5.1/8.4	This study
Average (b)	K	6	-	-	68.4	224.5	A_{95}=3.1	
Junggar								
Manas	K	22	44.2	86.0	73.4	223.6	A_{95}=4.3	[12,13]
Karamay IS	J2	11	45.6	84.9	74.8	215.9	4.2/6.0	This study
Karamay TC	J2	11	"	"	74.6	190.7	7.2/9.7	"
Eurasia	65-130Ma	29	-	-	76.3	202.6	A_{95}=2.2	[32]

S lat., S Long. = Average latitudes and longitudes of sampling areas. *P lat., P. Long.* = Latitudes and longitudes of VGPs, with d_p/d_m the angles of their cones of confidence.
(a) Previous estimate of the Tarim VGP from [12] based on five studies of K_1–K_u studies.
(b) New estimate of the Tarim VGP including the result for the Turfan basin from the present study.

puted on the basis of both in-situ and tilt-corrected directions (Table 2 and Fig. 6b), keeping in mind the fact that the tilting of the Middle Jurassic beds is likely to be of pre-Late Jurassic in age [15,17].

If the magnetization has been acquired by horizontal beds, it has a Middle Jurassic age, because of the age of tilting. We may see, in Fig. 6b, that this tilt-corrected pole together with the previously ob-

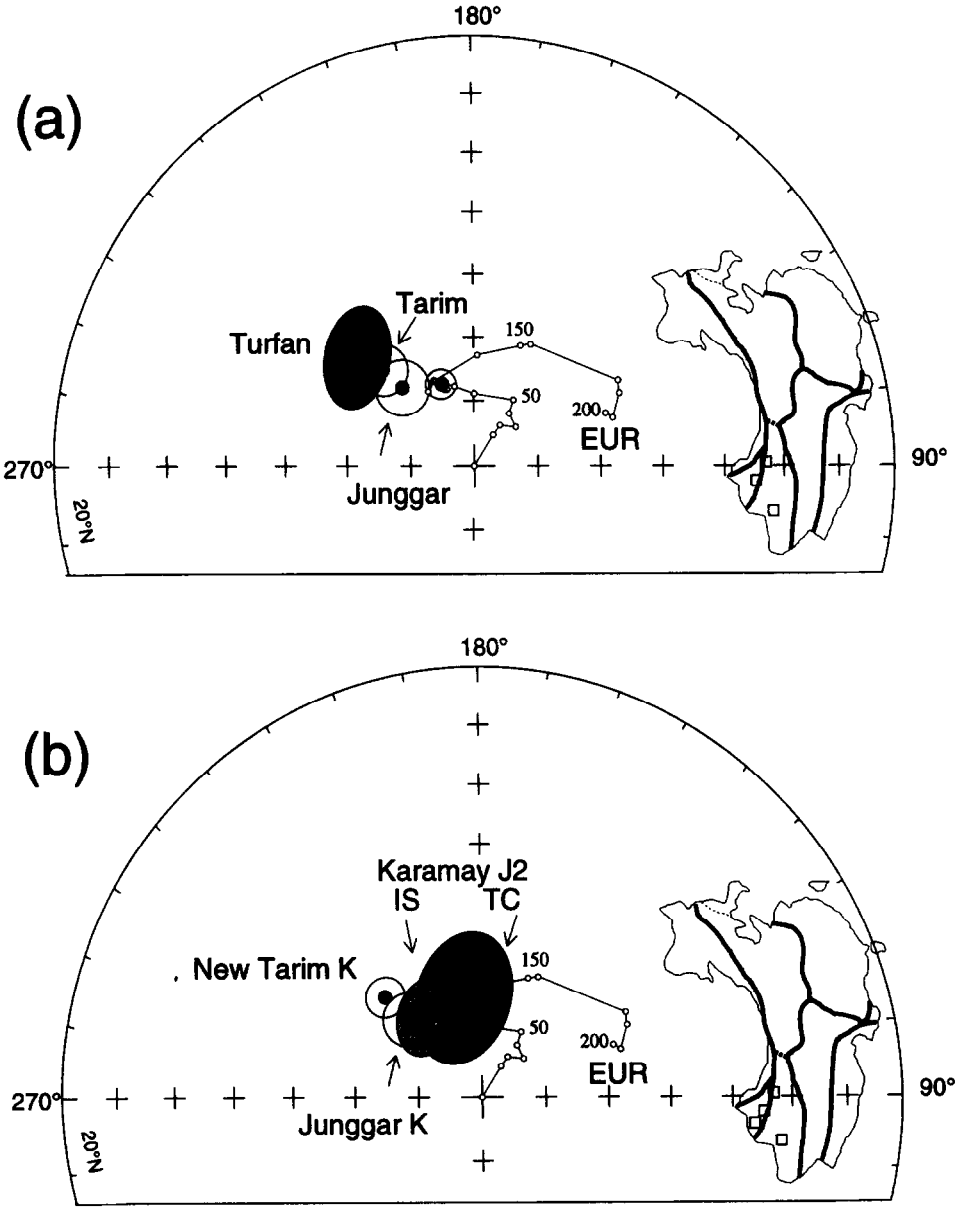


Fig. 6. (a) Equal-area projections on the northern hemisphere of the Eurasian APWP (age indicated in Ma) and the Cretaceous VGP of Turfan, Tarim, Junggar and Eurasia (*EUR*). (b) Equal-area projections on the northern hemisphere of the Eurasian APWP (age indicated in Ma), the Cretaceous VGP of Tarim (as computed in the present article) and Junggar, and the VGP of the Karamay area computed from tilt-corrected (TC, dark grey) and in-situ (IS, light grey) paleomagnetic directions obtained from Middle Jurassic J_{2s-x} Formations. See text and Table 2.

tained Cretaceous pole of Junggar [13] would define a small segment of APWP, compatible both with the Eurasian APWP and with the notion that convergence between Junggar and Siberia was accomplished only in the Tertiary, after the collision of India. If we now consider that the fold test is significantly negative we must discuss the pole derived from the in-situ data (Table 2 and Fig. 6b). Indeed, the in-situ directions lead to a better defined pole which is essentially identical to the Cretaceous (Upper Jurassic to Early Eocene) pole of Junggar, and significantly different from the present-day dipolar North Pole. Based on this observation, the beds may have been remagnetized in the Cretaceous, in a direction consistent with our previous interpretation of the Junggar data. However, in the absence of further fold or reversal tests, these conclusions are not robust, and we cannot conclude that we have been successful in determining a new Jurassic pole for the Junggar block.

Acknowledgements

This research was supported by the *Institut National des Sciences de l'Univers* (CNRS, France) and the Xinjiang Engineering Institute (People's Republic of China). This is IPGP contribution 1364. [*FA*]

References

[1] R.J. Enkin, Z. Yang, Y. Chen and V. Courtillot, Paleomagnetic constraints on the geodynamic history of the major blocks of China from the Permian to the Present, J. Geophys. Res. 97, 13953–13989, 1992.

[2] P. Molnar and P. Tapponnier, Cenozoic tectonics of Asia: effects of a continental collision, Science 189, 419–426, 1975.

[3] P. Tapponnier, G. Peltzer and R. Armijo, On the mechanics of the collision between India and Asia, Geol. Soc. London Spec. Publ. 19, 115–157, 1986.

[4] Y. Chen, V. Courtillot, J.-P. Cogné, J. Besse, Z. Yang and R. Enkin, The configuration of Asia prior to the collision of India: Cretaceous paleomagnetic constraints, J. Geophys. Res. 98, 21927–21941, 1993b.

[5] Z. Yang and J. Besse, Paleomagnetic study of Permian and Mesozoic sedimentary rocks from Northern Thailand supports the extrusion model for Indochina, Earth Planet. Sci. Lett. 117, 525–552, 1993.

[6] J.-P. Pozzi, M. Westphal, Y. Zhou, L. Xing and X. Chen, Position of the Lhasa block, South Tibet, during the late Cretaceous, Nature 297, 319–321, 1982.

[7] J. Achache, V. Courtillot and Y.X. Zhou, Paleogeographic and tectonic evolution of South Tibet since Middle Cretaceous time: new paleomagnetic data and synthesis, J. Geophys. Res. 98, 10331–10339, 1984.

[8] J.L. Lin and D.R. Watts, Paleomagnetic results from the Tibetan Plateau, Philos. Trans. R. Soc. London A327, 239–262, 1988.

[9] Y. Otofuji, S. Funahara, J. Matsuo, F. Murata, T. Nishiyama, X. Zheng and K. Yaskawa, Paleomagnetic study of western Tibet: Deformation of a narrow zone along the Indus Zangbo suture between India and Asia, Earth Planet. Sci. Lett. 92, 307–316, 1989.

[10] Y. Chen, J.P. Cogné, V. Courtillot, P. Tapponnier and X.Y. Zhou, Cretaceous paleomagnetic results from Western Tibet and tectonic implications, J. Geophys. Res. 98, 17981–18000, 1993a.

[11] Y.P. Li, Z.K. Zhang, M. McWilliams, R. Sharps, Y.J. Zhai, Y.A. Li, Q. Li and A. Cox, Mesozoic paleomagnetic results of the Tarim craton: Tertiary relative motion between China and Siberia?, Geophys. Res. Lett. 15, 217–220, 1988.

[12] Y. Chen, J.-P. Cogné and V. Courtillot, New Cretaceous paleomagnetic results from the Tarim basin, Northwestern China, Earth Planet. Sci. Lett. 114, 17–38, 1992.

[13] Y. Chen, J.P. Cogné, V. Courtillot, J.P. Avouac, P. Tapponnier, E. Buffetaut, G.Q. Wang, M.X. Bai, H.Z. You, M. Li and C.S. Wei, Paleomagnetic study of Mesozoic continental sediments along the Northern Tien Shan (China) and heterogeneous strain in central Asia, J. Geophys. Res. 96, 4065–4082, 1991.

[14] X. Peng and G. Zhang, Tectonic features of the Junggar basin and their relationship with oil and gas distribution, in: Chinese Sedimentary Basins (Sedimentary Basins of the World), pp. 17–31, Elsevier, 1989.

[15] Q. Wu, Structural evolution and prospects in the Junggar basin, Xinjiang Geol. 4, 20–34, 1986 (in Chinese, with English Abstr.).

[16] A.W. Bally, I. Ming Chou, R. Clayton, H.P. Heugster, S. Kidwell, L.D. Meckel, R.T. Ryder, A.B. Watts and A.A. Wilson, Note on sedimentary basins in China. Report of the American Sedimentary Basins Delegation to the People's Republic of China, August 17–September 8, 1985, U.S. Geol. Surv. Open-File Rep. 86-327, 108 pp., 1986.

[17] J.-P. Avouac, P. Tapponnier, M. Bai, H. You and G. Wang, Active thrusting and folding along the Northern Tien Shan and Late Cenozoic rotation of the Tarim relative to Dzungaria and Kazakhstan, J. Geophys. Res. 98, 6755–6804, 1993.

[18] J.D.A. Zijderveld, A.C. demagnetization of rocks: Analysis of results, in: Methods in Paleomagnetism, D.W. Collinson, K.M. Creer and S.K. Runcorn, eds., pp 254–286, Elsevier, 1967.

[19] J.L. Kirschvink, The least-squares line and plane and the analysis of paleomagnetic data, Geophys. J. R. Astron. Soc. 62, 699–718, 1980.

[20] R.A. Fisher, Dispersion on a sphere, Proc. R. Soc. London A217, 295–305, 1953.
[21] P.L. McFadden and M.W. McElhinny, The combined analysis of remagnetization circles and direct observations in paleomagnetism, Earth Planet. Sci. Lett. 87, 161–172, 1988.
[22] K.V. Mardia, Statistics of Directional Data, Academic Press, 1972.
[23] H.C. Halls, The use of converging remagnetization circles in palaeomagnetism, Phys. Earth Planet. Inter. 16, 1–11, 1978.
[24] M.W. McElhinny, Statistical significance of the fold test in paleomagnetism, Geophys. J. R. Astron. Soc. 8, 338–340, 1964.
[25] P.L. McFadden and F.J. Lowes, The discrimination of mean directions drawn from Fisher distributions, Geophys. J. R. Astron. Soc. 67, 19–33, 1981.
[26] M.J. Dekkers, Magnetic properties of natural pyrrhotite. I: Behaviour of initial susceptibility and saturation-magnetization-related rock-magnetic parameters in a grain-size dependent framework, Phys. Earth. Planet. Inter. 52, 376–393, 1988.
[27] M.J. Dekkers, Magnetic properties of natural pyrrhotite. II: High-and low-temperature behaviour of J_{rs} and TRM as a function of grain size, Phys. Earth. Planet. Inter. 57, 266–283, 1989.
[28] C. Mary, S. Iaccarino, V. Courtillot, J. Besse and D.M. Aissaoui, Magnetostratigraphy of Pliocene sediments from the Stirone River (Po Valley), Geophys. J. Int. 112, 359–380, 1993.
[29] W. O'Reilly, Rock and Mineral Magnetism, Chapman&Hall, 1984.
[30] R. Lovlie, G. Stole and N. Spjeldnaes, Magnetic polarity stratigraphy of Plio-Pleistocene marine sediments from Rhodos, eastern Mediterranean, Phys. Earth. Planet. Inter. 54, 340–352, 1989.
[31] J.-P. Cogné, N. Bonhommet, V. Kropacek, T. Zelinka and E. Petrovski, Paleomagnetism and magnetic fabric of the deformed redbeds of the Cap de la Chèvre formation, Brittany, France, Phys. Earth Planet. Inter. 67, 374–388, 1991.
[32] J. Besse and V. Courtillot, Revised and synthetic polar wander paths of the African, Eurasian, North American and Indian plates, and true polar wander since 200 Ma, J. Geophys. Res. 96, 4029–4050, 1991.
[33] V. Courtillot, R. Enkin, Z. Yang, Y. Chen, M. Bazhenov, J. Besse, J.P. Cogné, R. Coe, X. Zhao and S. Gilder, Response to a comment on 'Paleomagnetic constraints on the geodynamic history of the major blocks of China from the Permian to the Present' J. Geophys. Res., in press, 1994.
[34] L. Tauxe and G.S. Watson, The fold test: an eigen analysis approach, Earth Planet. Sci. Lett. 122, 331–342, 1994.
[35] P.L. McFadden and D.L. Jones, The fold test in paleomagnetism, Geophys. J. R. Astron. Soc. 67, 53–58, 1981.